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MILITARY STANDARD

FASTENER TEST METHODS

METHOD 8,

TENSILE STRENGTH

FSC 53GP

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DEPARTMENT OF DEFENSE
WASHINGTON, DC 20301

Fastener Test Methods, Method 8, Tensile Strength

MIL-STD-1312-8A

1. This Military Standard is approved for use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD), Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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FOREWORD

This standard sets forth a standard test method to determine the tensile strength of externally and internally threaded fasteners.

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TABLE OF CONTENTS

Paragraph		Page
1.	SCOPE	1
1.1	Applicability	1
2.	REFERENCED DOCUMENTS	1
2.1	Government documents	1
2.1.1	Specifications, standards and handbooks	1
2.2	Other publications	1
3.	DEFINITIONS	2
4.	GENERAL REQUIREMENTS	2
4.1	Test apparatus	2
4.1.1	Testing machine	2
4.1.2	Extension measuring device	2
4.1.3	Test fixtures	2
4.1.3.1	NAS 1069 fixture	3
4.1.3.2	Tension plate fixture	3
4.1.3.3	Angularity evaluation	3
4.2	Test specimens	3
4.2.1	General	3
4.2.2	Externally threaded fasteners	3
4.2.3	Internally threaded fasteners	3
4.2.4	Grip length	3
4.2.4.1	Specified grip range	3
4.2.4.2	Unspecified grip range	8
4.2.4.3	Grip length adjustment	8
4.2.5	Washers or spacers	8
5.	DETAIL REQUIREMENTS	8
5.1	Test procedures	8
5.1.1	Test setup	8
5.1.1.1	NAS 1069 fixture	8
5.1.1.2	Tension plate fixture	8
5.2	Loading rate	8
5.3	Load-deflection curves	9
5.4	Failing load determination	9
5.5	Yield load determination	9
5.6	Number of specimens	9
6.	NOTES	12
6.1	Test report	12

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FIGURES

Figure		Page
1.	Tension fixtures for short grip fasteners	4
2.	Tension setup	5
3.	Test plate	6
4.	Tension fixture assembly	7
5.	Sample load-deflection curve	10
6.	Johnson's two-thirds approximate method for determination of yield strength	11

TABLE

Table		Page
I.	Tension load rates	9

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1. SCOPE

1.1 Applicability. This method covers a procedure for testing all types of structural fasteners in tension at room temperature. This procedure is intended to define the test required to determine the tensile strength of the fastener itself, not the strength of the fastener in any sheet combination.

2. REFERENCED DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. Unless otherwise specified, the following specifications, standards and handbooks of the issue listed in the current Department of Defense Index of Specifications and Standards (DoDISS) and the supplement thereto (if applicable), form a part of this standard to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-45662 Calibration System Requirements

(Copies of specifications, standards, handbooks, drawings and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) forms a part of this specification to the extent specified herein. The issues of the documents which are indicated as DOD adopted shall be the issue in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E4 Standard Methods of Verification of Testing Machines

ASTM E 83 Tentative Method of Verification and Classification of Extensometers

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AEROSPACE INDUSTRIES ASSOCIATION, INC. (AIA)

NAS 1069 Tension Fatigue Test Procedure for Aeronautical Fasteners

(Application for copies should be addressed to the Aerospace Industries Association, Inc., 1725 DeSales Street, NW, Washington, DC 20036.)

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3. DEFINITIONS

Not applicable.

4. GENERAL REQUIREMENTS

4.1 Test apparatus.

4.1.1 Testing machine. The testing machine shall be capable of applying a tensile load at a controllable rate. The calibration system for the machine shall conform to the requirements of MIL-STD-45662. Its accuracy shall be verified every 12 months by a method complying with ASTM E 4, using a calibration device that shall have been calibrated by the National Bureau of Standards not more than 2 years prior to its use. The yield loads, ultimate loads, and structural failure loads of the fasteners tested (5.4 and 5.5) shall be within the loading range of the testing machine as defined in ASTM E 4.

4.1.2 Extension measuring device. The extension measuring device shall be an averaging, differential-transformer extensometer, or equivalent, preferably of the separable type. It shall conform to the requirements of ASTM E 83, Class B1 when used in conjunction with an autographic recorder. The extensometer shall be capable of installation so as to measure either the relative movement between the movable and stationary crossheads of the testing machine or the extension of the fastener only, preferably the latter. Load and extension ranges shall be used to give the initial portion of the load-extension curve a slope between 45 and 60 degrees.

4.1.3 Test fixtures. Test fixtures specified in 4.1.3.1 and 4.1.3.2 are permissible. Other types of fixtures may be used provided all of the following requirements are met:

a. Hole size for any fastener shall be the maximum shank diameter of the fastener to be tested plus 0.001-inch with a tolerance of +0.004, -0.000-inch. Alternatively, the hole shall conform to the diameter and limits specified by the fastener manufacturer or applicable procurement specification.

b. The hole shall be perpendicular to the fixture surfaces within ±1 degree.

c. Where applicable, the hole shall be chamfered to provide clearance for the head-to-shank fillet of the fastener.

d. The fixture shall be capable of applying an axial tensile load through the center line of the fastener by means of suitable supporting fixtures.

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4.1.3.1 NAS 1069 fixture. This recommended test fixture is the cup or insert type as defined in NAS 1069, figures 8-13. Hole size for any unspecified fastener shall be the maximum shank diameter of the fastener to be tested plus 0.001-inch with a +0.004, -0.000-inch tolerance. In any case, the hole in the fixture shall not reduce the bearing area by more than 10 percent. The bearing area shall be considered as the area between minimum bearing face outer diameter and the maximum diameter at the point of tangency of the head-to-shank fillet. The fixture design shall provide clearance for the shank-to-head fillet.

4.1.3.2 Tension plate fixture. The tension plate type fixture may be used for short grip length fasteners. The fixture configuration shall be as shown on figures 1 and 2 or figures 3 and 4. The upper and lower plate shall be parallel within 0.25 degree.

4.1.3.3 Angularity evaluation. When tests evaluate the effect of angularity under the head or nut (collar) end of the fastener, the angularity shall be applied by the use of tapered spacers of the required angularity. The spacers shall be made of material harder than that of the fastener and ground flat. The outside diameter shall be greater than the head size of the fastener. The hole shall be the maximum shank diameter plus 0.005-inch with a tolerance of +0.003, -0.000-inch. Alternatively, the hole shall conform to the diameter and limits specified by the fastener manufacturer or applicable government specification.

4.2 Test specimens.

4.2.1 General. When specific bolt-nut combinations, pin and collar fasteners, or integral fasteners supplied complete in one assembly, are being evaluated, the tests must be performed on the entire assembly. Tests resulting in failure of any component of the assembly shall be considered a satisfactory test of the fastener.

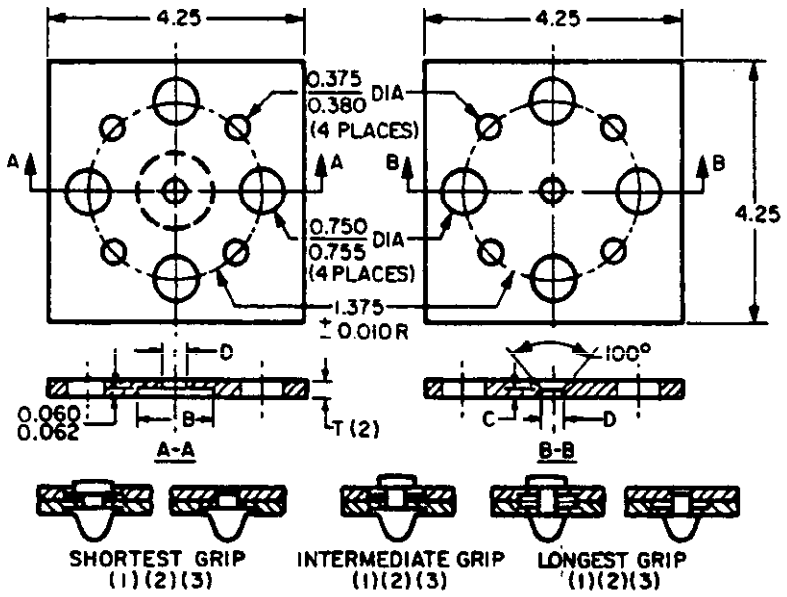
4.2.2 Externally threaded fasteners. The nut used for testing of the externally threaded fastener alone shall be of sufficient strength to ensure bolt failure. The nut portion can be an integral part of the test fixture. Tests resulting in nut failure shall not be considered a satisfactory test of the bolt.

4.2.3 Internally threaded fasteners. The bolt used for testing of the internally threaded fastener alone shall be of sufficient strength to ensure nut failure. Tests resulting in bolt failure shall not be considered a satisfactory test of the nut.

4.2.4 Grip length.

4.2.4.1 Specified grip range. Fasteners having a specific range (such as lockbolts, HI-LOKS, blind rivets, and blind bolts) shall be tested in such grip conditions as required by applicable specifications or procurement documents.

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NOTES:

1. Omit counterbore and use 2024-T3 or 2024-T4 aluminum sheet when testing DEUTSCH blind rivets, unless otherwise specified.
2. Counterbore optional, if short and intermediate grips are not required.
3. Surface finish R_a 32 to 63.
4. Material: 4140 alloy steel heat treated to 200,000 psi, or equal (1).
5. Tolerances: ±0.010-inch and ±1°, unless otherwise indicated.
6. Chamfer all hole edges to provide head fillet clearance.
7. All dimensions in inches.

Nom fstnr size	T +0.001 -0.000 (in.)	B dia ±0.01 (in.)	D diameter, +0.004, -0.000 (in.)						C dimension, +0.000, -0.002 (in.)						
			(a,b, c,d)	(e)	(f)	(g)	(h)	(j)	(a)	(b,h)	(c,j)	(d)	(e)	(f)	(g)
0.093	0.125	0.75	0.094				0.097			0.036					
0.125	0.125	0.75	0.125			0.136	0.129			0.042		0.030			0.039
0.156	0.188	0.88	0.156	0.164	0.170	0.173	0.160			0.055		0.038	0.072	0.070	0.049
0.164	0.188	0.88	0.165						0.072						
0.190	0.250	1.00	0.190	0.199	0.197	0.204	0.192	0.190	0.084	0.070	0.048	0.046	0.084	0.080	0.064
0.190	0.250	1.00	0.190	0.199	0.197				0.084	0.070	0.048		0.084	0.080	
0.250	0.250	1.00	0.250	0.261	0.258		0.256	0.260	0.110	0.095	0.063	0.061	0.111	0.108	
0.313	0.250	1.12	0.312	0.313	0.339				0.138	0.106	0.070		0.139	0.128	
0.375	0.250	1.12	0.375	0.375	0.391				0.159	0.134	0.081		0.167	0.160	

(a) Protruding and flush tension head solid shank fasteners
(b) Flush MS20426 head solid shank fasteners
(c) Flush head solid shank fasteners
(d) Flush shear head driven rivets
(e) JO-bolts
(f) Blind bolts
(h) Blind rivets except DEUTSCH
(j) DEUTSCH blind rivets (see note 1)

FIGURE 1. Tension fixtures for short grip fasteners.

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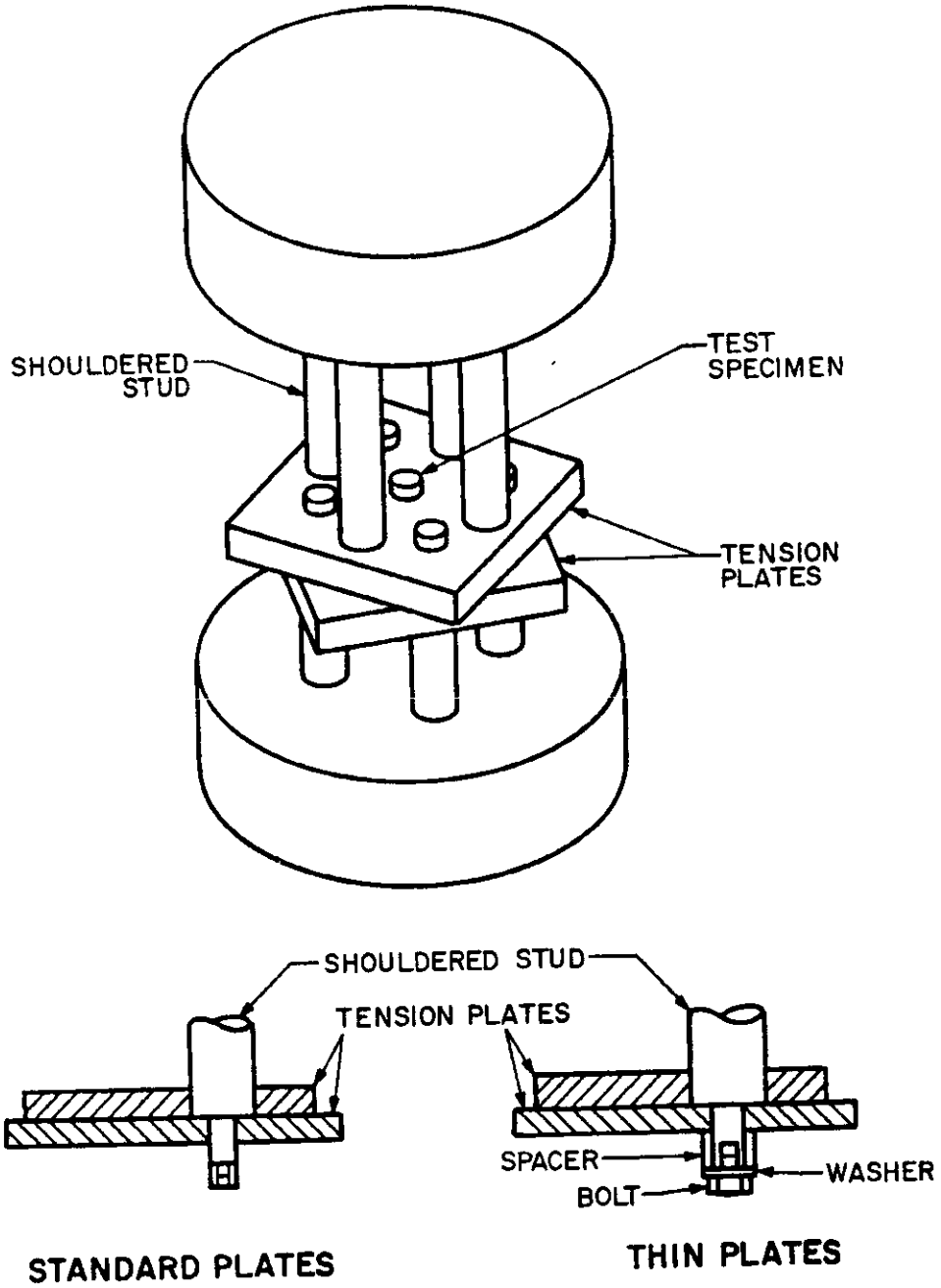
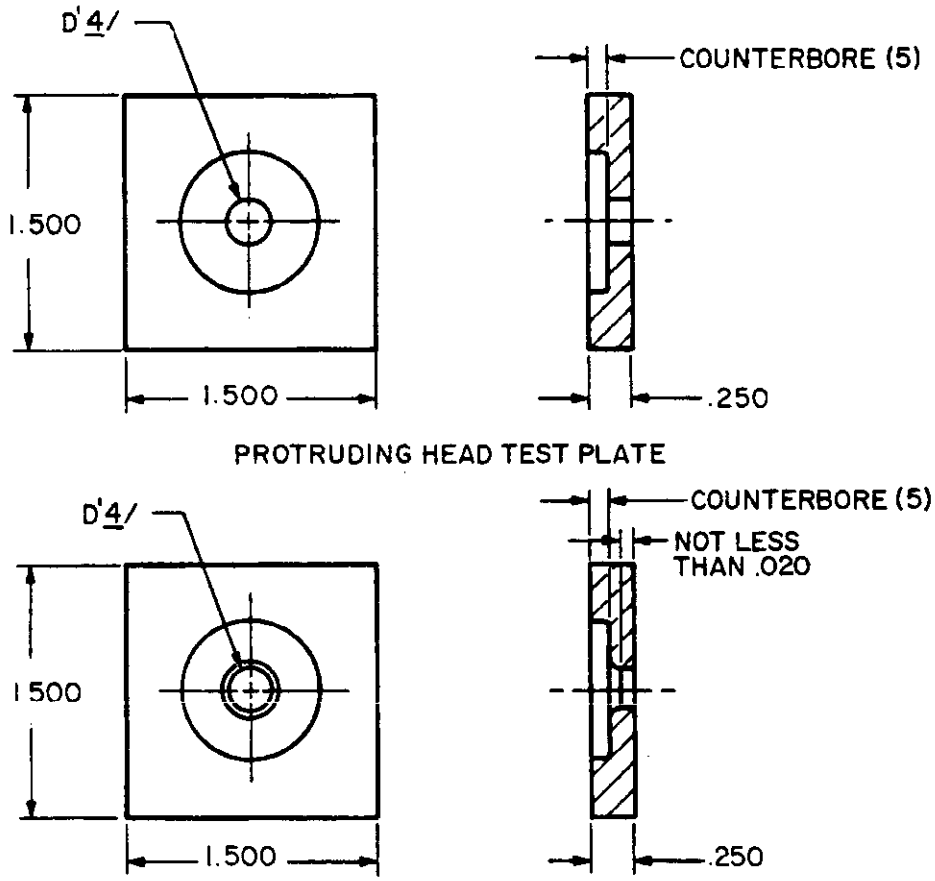


FIGURE 2. Tension setup.

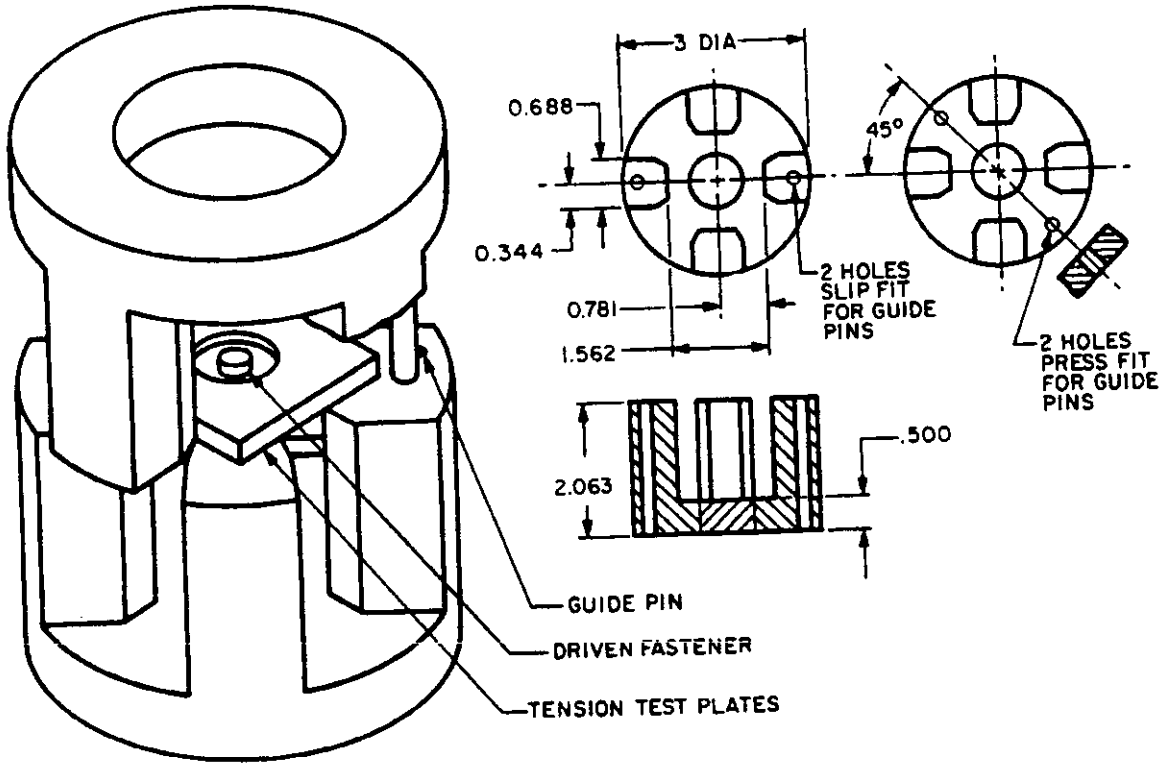
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NOTES:
1. Material: Alloy steel.
2. Heat treat: 48-50 HRC.
3. Machine or grind edges square with face.
4. D' = (D +0.001), +0.002, -0.000, where D equals the fastener maximum shank diameter.
5. Counterbore as required for grip.
6. All dimensions in inches.

FIGURE 3. Test plate.

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NOTES:
1. All dimensions in inches.

FIGURE 4. Tension fixture assembly.

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4.2.4.2 Unspecified grip range. Threaded fasteners having no specified grip range shall be tested at the grip length where two to three bolt threads extend below the face of the nut. The bolt chamfer shall extend beyond the top of the nut.

4.2.4.3 Grip length adjustment. The grip length for fasteners which must be installed with installation preload shall be adjusted with washers or shims located between the test fixtures, not under nuts, collars or heads.

4.2.5 Washers or spacers. The use of washers under fastener heads, nuts or collars is specifically prohibited unless the use of such washers is specifically required in the product application. Where spacers are required to evaluate the effect of angularity or to provide clearance for the head fillet, they shall be hardened and ground as specified in 4.1.3.3.

5. DETAIL REQUIREMENTS

5.1 Test procedures.

5.1.1 Test setup.

5.1.1.1 NAS 1069 fixture. The specimen shall be installed in the holding fixture at the proper grip using normal installation procedures. Preload shall not be used except for those fasteners where preload is a normal result of the installation. The assembly of specimen and fixtures shall be placed between the machine tension heads. The difference between the maximum and the average stress caused by misalignment or eccentric loading shall not exceed 3 percent average stress.

5.1.1.2 Tension plate fixture. The specimens shall be installed in the fixture at the proper grip using normal installation procedures. Preload shall not be used except for those fasteners where preload is a normal result of the installation. The assembly of specimen and plates shall be installed in a suitable test jig as shown on figures 2 or 4 and placed between the compression heads of the testing machine. Care shall be exercised to locate the jig at the center of the piston when hydraulic testing machines are used.

5.2 Loading rate. Tension load shall be applied to the fastener slowly and evenly. Unless otherwise specified by the fastener manufacturer, or by applicable detail specification or procurement document, the rate of loading shall be in accordance with table I. Larger or smaller fasteners than those shown in table I shall be tested at rates producing 100,000 pounds/minute (+10 percent) per square inch of nominal shank area.

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TABLE I. Tension load rates.

Nominal fastener diameter	Load rate (lb/min)	Nominal fastener diameter	Load rate (lb/min)
0.125	1,240	0.563	24,800
0.156	1,920	0.625	30,600
0.164	2,100	0.750	44,000
0.190	2,800	0.875	60,000
0.250	5,000	1.000	78,000
0.313	7,700	1.125	100,000
0.375	11,000	1.250	122,000
0.438	15,000	1.375	148,000
0.500	19,600	1.500	176,000

NOTE: As an alternative loading method, the rate of loading can be determined as a constant head travel that gives strain rates consistent with the above load rates in the elastic range.

5.3 Load-deflection curves. Load-deflection curves shall be made by autographic recording. The movement sensing element shall be so installed as to either measure the relative movement between the movable crosshead and the stationary crosshead, or to measure the deflection of the fastener only, preferably the latter. The latter measurement shall be accomplished by placing the movement sensing elements adjacent to the fastener between the facing surfaces of the fastener holding fixtures for the NAS 1069 fixture, or on the externally opposed surfaces of the tension plate, or over the ends of the fastener. The fixture shall have the capability of being non-yielding at ultimate load.

5.4 Failing load determination. When a structure failure occurs before the ultimate or fracture load, it may be detected on the load-deflection curve as a peak load followed by severe permanent deformation without increase in load or with a decrease in load. A sample load-deflection curve illustrating a "structural failure" is shown on figure 5. The first peak load shall be designated as the "structural failure load" and the second peak shall be the "ultimate load."

5.5 Yield load determination. The yield load shall be determined by the Johnson's two-thirds approximate method, as shown on figure 6. On the load-deflection plot of 5.3, draw a line (line A) with two-thirds the slope of the straight portion of the curve. Parallel to this line and tangent to the load-deflection curve, draw another line (line B). The point of tangency of line B with the curve represents the yield load.

5.6 Number of specimens. Unless otherwise specified in the detail specification or purchase order, a minimum of five specimens shall be tested for each fastener diameter and test condition.

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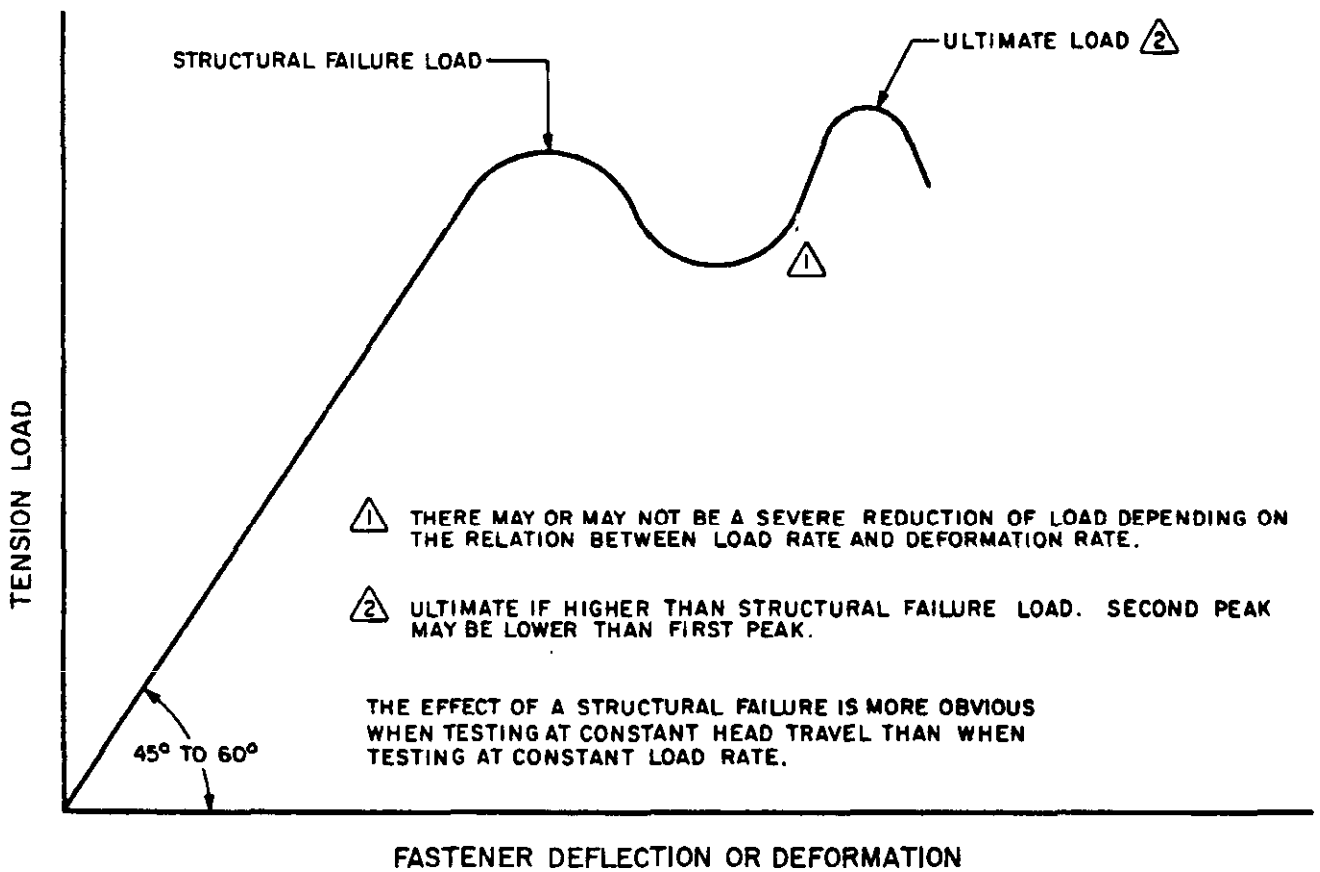


FIGURE 5. Sample load-deflection curve.

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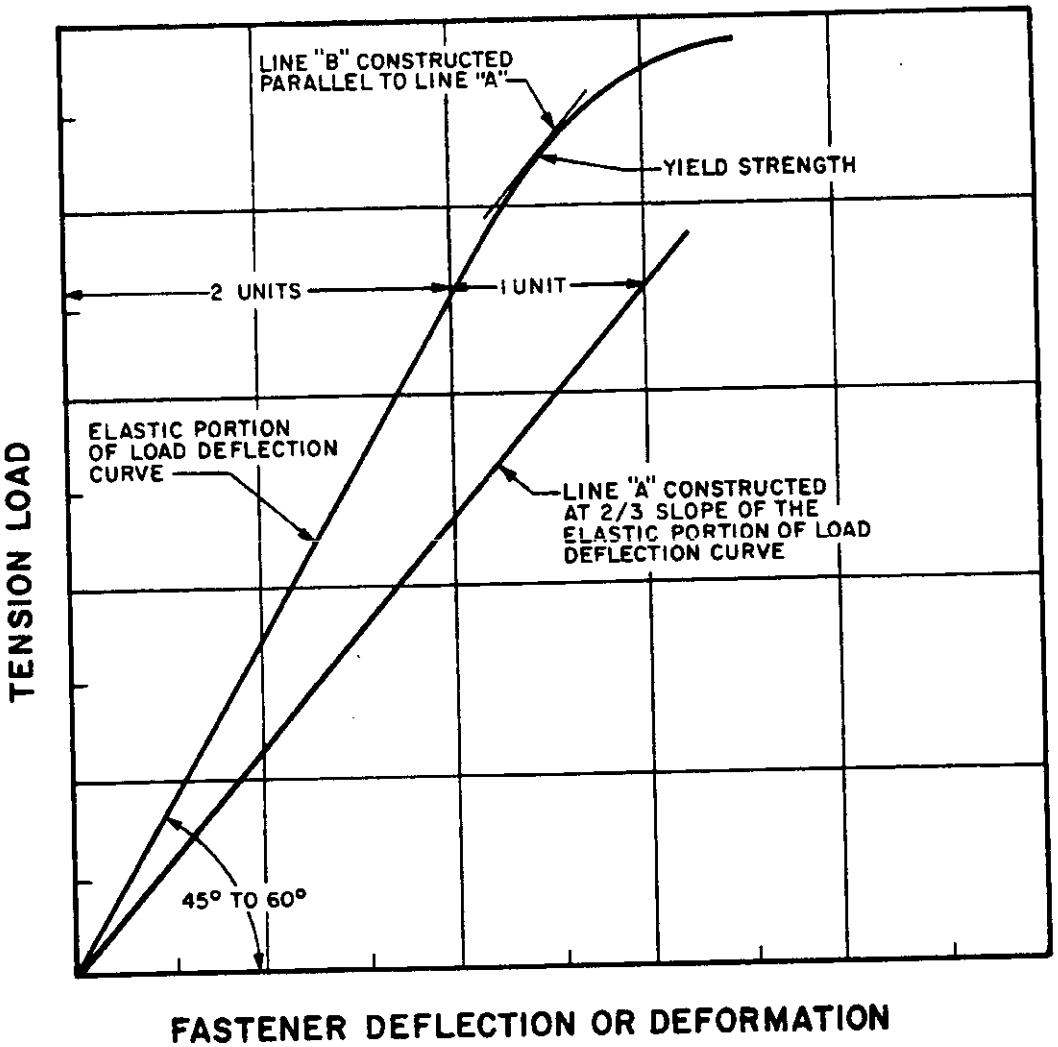


FIGURE 6. Johnson's two-thirds approximate method for determination of yield strength.

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6. NOTES

6.1 Test report. The test report shall contain the following:

a. Fastener description.

1. Part number.

2. Lot identification.

3. Manufacturer.

4. Material.

5. Heat treatment level.

6. Measured shank diameters.

7. Grip length.

b. Test machine.

1. Model and serial number.

2. Calibration date.

c. Loading rate or strain rate.

d. Ultimate loads.

e. Yield loads.

f. Installation procedure.

g. Fixture used.

h. Structural failure loads.

i. Mode of failure.

j. Load-deflection curve.

k. Strain magnification factor.

l. Method of deflection measurement.

m. Cause and duration of any interruptions during test.

n. Results of all inspections.

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